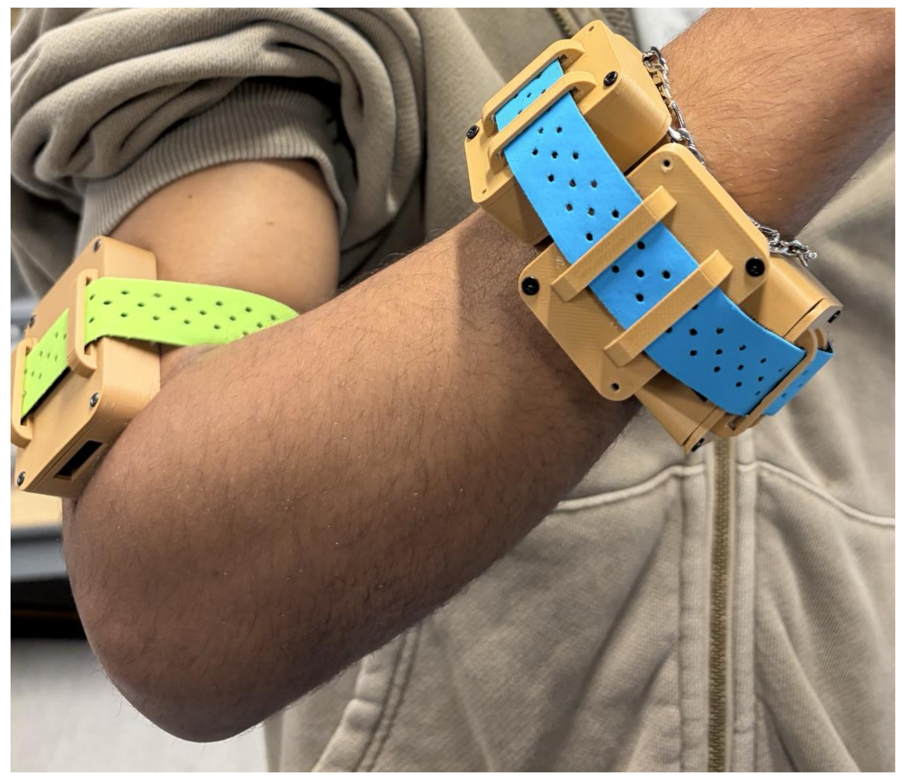

FlexType EMG Keyboard

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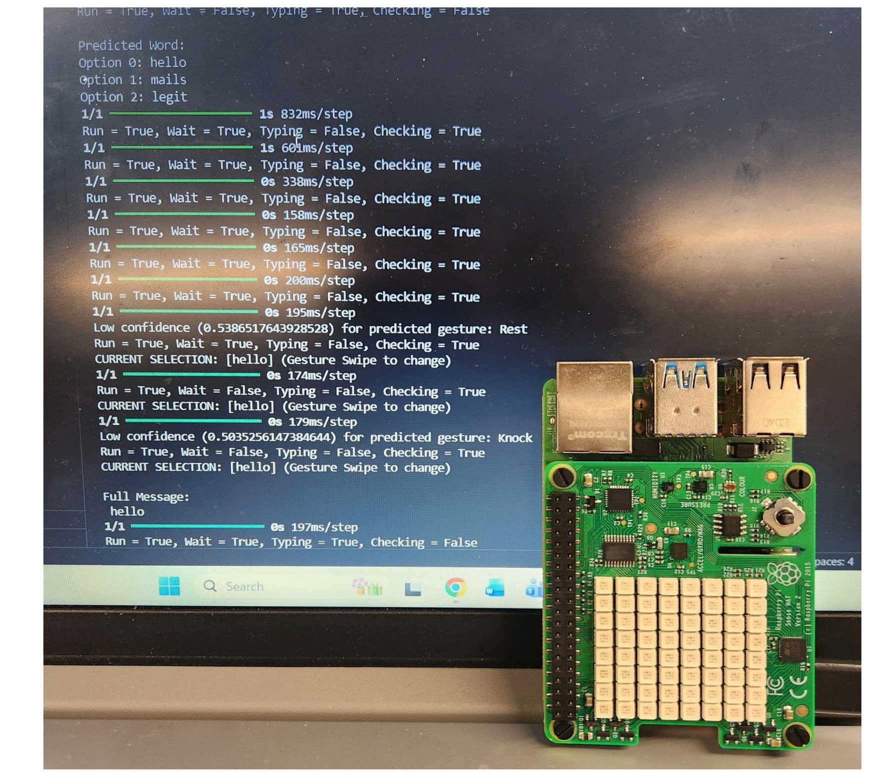


Introduction

People with fine motor control issues can struggle using traditional keyboards and touchscreens. With their limited control over their hand movement, it can be hard to accurately tap on a specific location. With the FlexType EMG Keyboard that is no longer the case.

This wearable wristband can read the electromyographic (EMG) signals in your muscles, removing the need to physically touch a screen or keyboard. Instead, the user performs a gesture, and a machine learning algorithm will recognize it. It then uses that information along with a second integrated machine learning algorithm to predict the word being typed.

Market Research

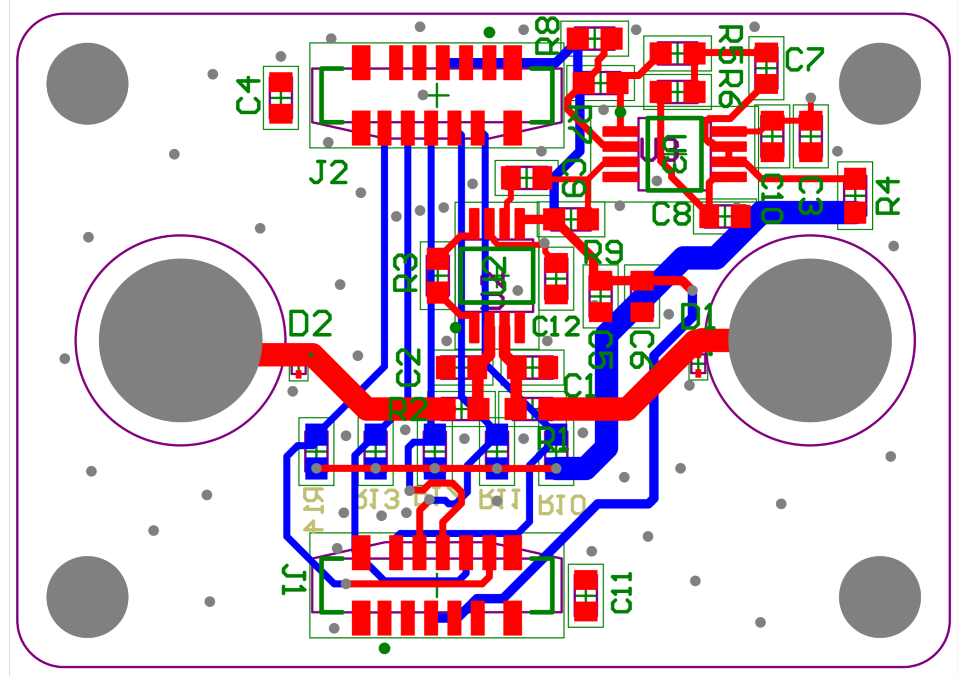
- The FlexType EMG Keyboard sits between the gesture recognition market valued at $25.4B and the wearable EMG market.
- Our main audience deals with people with limited motor control, as well as the elderly, visually impaired, and people missing limbs. This market can potentially be made up of 50-60 million people in the U.S.
- Our biggest competitor is Meta, with their VR/AR controlled wristband they are currently developing.

Project Requirements

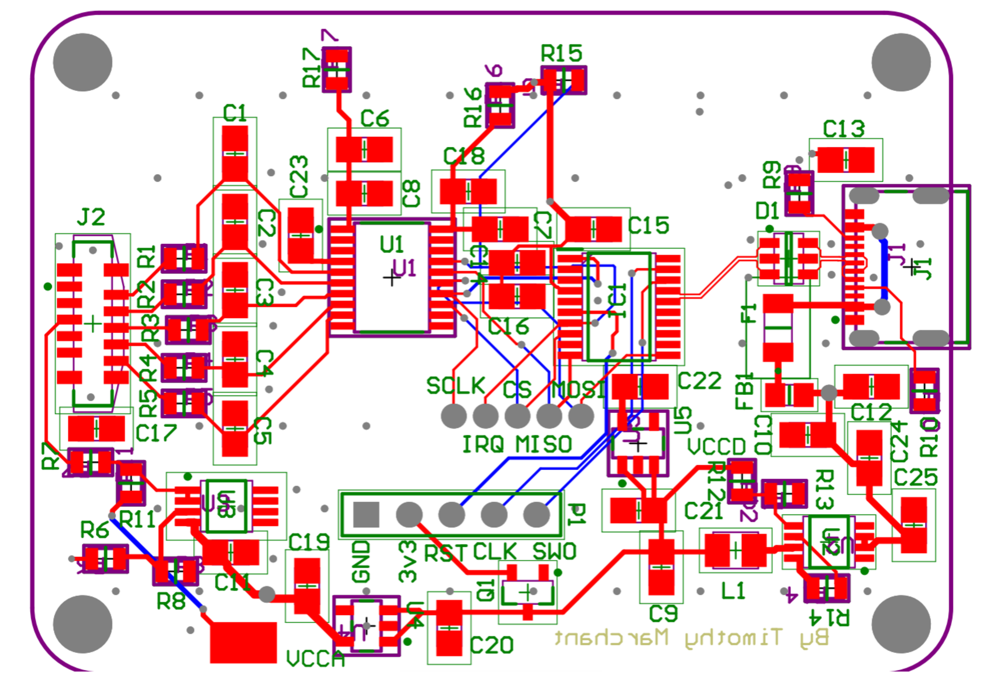
- The FlexType EMG wristband subsystem shall capture and digitize EMG signals from the user's forearm into readable data for processing
- The FlexType model subsystem shall interpret EMG data to classify gestures and generate predicted text output using machine learning
- Shall display the predicted text in real time and allow for user interactions, like corrections and selections.
- The case should enclose the entire device, have openings for wire connectors and electrodes.
- The case shall be as small as possible so that multiple PCBs fit on the user arm.

Design

Electrode Board



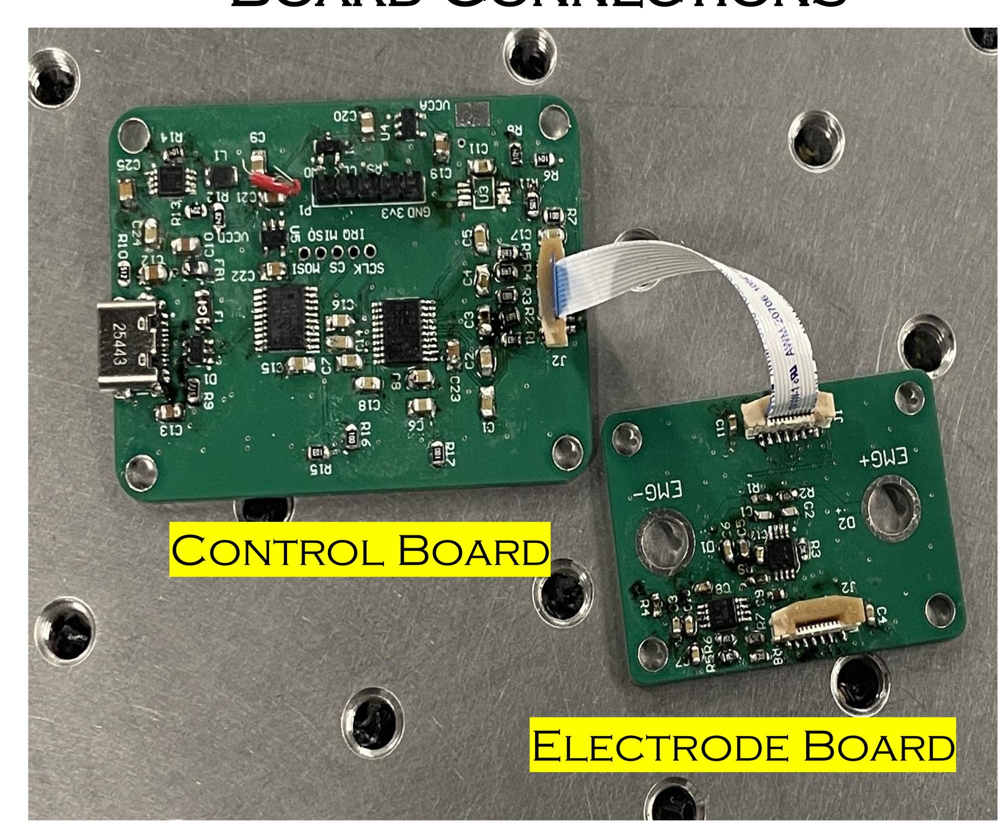
Control Board

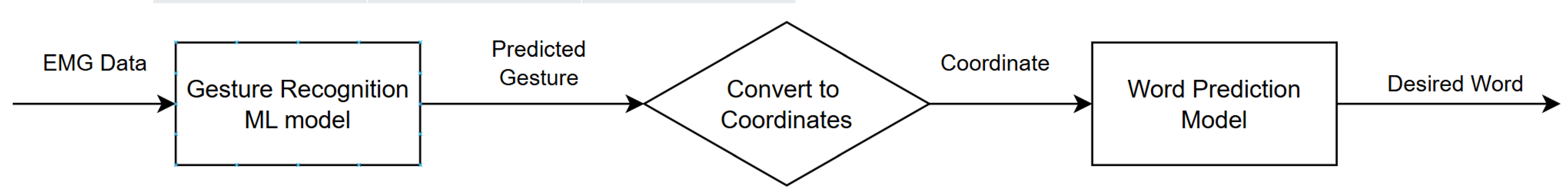


Gestures & Commands

Gesture	Command	Prediction Accuracy
Rest	-	98.65%
Middle Pinch	A-F	81.74%
Ring Pinch	G-M	96.94%
Pinky Pinch	M-T	85.26%
Three Fingers	U-Z	84.21%
L-Sign	Swipe	99.14%
Thumb-Out	Delete	92.88%
Knock	Space/ Enter	98.69%
Surfs Up	Start/ Stop	86.90%
	Overall:	93.08%

Board Connections





Results

•Challenges:
- Components were delayed pushing back assembly of PCBs
- FFC connectors were designed with incorrect orientation
- Word prediction model could not be run locally on the Raspberry Pi
- We were not able to get many volunteers for data collection

•Successes:
- Achieved 93% accuracy on gesture prediction model
- Successfully implemented remote connection to FlexType model
- PCB can successfully capture live EMG signals from muscle movements

Design Details

- **Hardware Implementation:**
  - A control board designed with 0805 SMD passives and EMG Electrode board with 0603.
  - Boards connect via 10-pin 0.5mm pitch FFC cables.
  - Dry electrodes using binding screws mounted directly through the PCB
  - STM microcontroller that is a SPI to USB bridge, reading the digital EMG data from the ADC and streaming it over USB-C to a Raspberry Pi
  - Custom enclosures modeled in Fusion 360
  - Two-part design: body and lid secured together with M2 & M3 screws, respectively .
  - Contains cutouts for USB-C port, FFC connectors, and electrode mounting points.

- **Software Implementation:**
  - Built custom EMG dataset with volunteers
  - Designed and implemented a 3-layer 1D CNN gesture prediction model using TensorFlow
  - Assigned gestures to different letter groups and commands
  - Integrated pre-existing FlexType model to convert gestures into predicted words

Bill of Materials

ID	Item	Quantity	Cost
01	MCP3464R ADC	2	$9.82
02	STM32C071FBP6 MCU	2	$2.80
03	INA333	5	$22.70
04	Op-Amp	5	$20.35
05	Resistors/Capacitors	230	$16.13
06	Screw Type Electrodes	1 set	$9.99
07	FFC Cables (50/100/150mm)	15	$45.30
08	Other Components (Passives, Connectors, etc)	--	$53.05
09	Raspberry Pi 4	1	$67.95
		Total Cost:	**$248.09**